

RoboSax Melody Slot Machine

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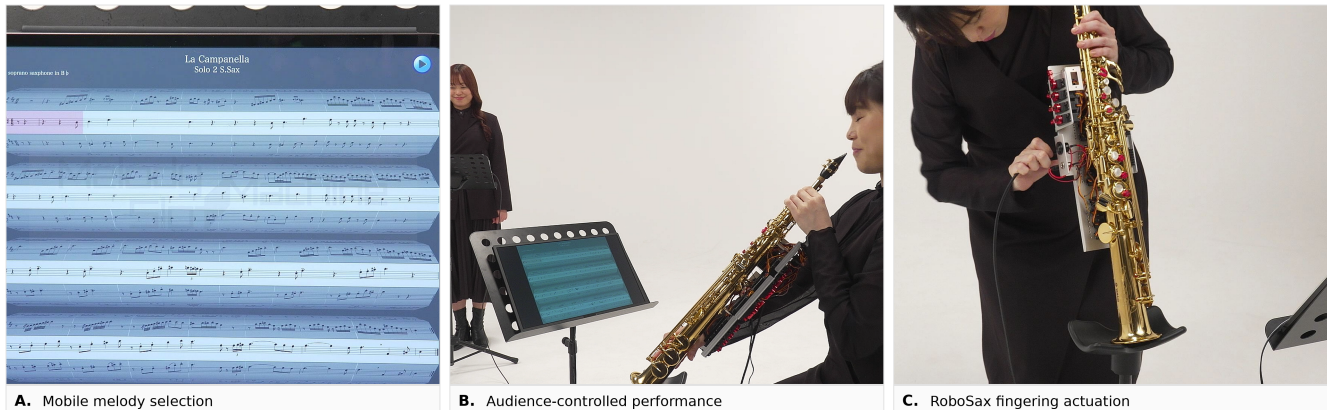


Figure 1: RoboSax Melody Slot Machine as a mobile-to-acoustic performance experience. (A) The Melody Slot Machine interface presents structurally related melodic fragments on a tablet, allowing participants to select variations through a playful slot-machine-like interaction grounded in GTTM-based melody morphing. (B) The selected melody becomes part of a live performance, enabling audience members to influence the musical structure without requiring instrumental training. (C) RoboSax physically actuates saxophone fingerings from MIDI messages, while the human performer provides breath, tonguing, phrasing, dynamics, and tone color.

Abstract

RoboSax Melody Slot Machine is a mobile-to-acoustic performance system that connects an interactive tablet interface to live saxophone performance. Participants select melodic fragments on an iPad-based Melody Slot Machine, where related musical phrases are generated through GTTM-based melody morphing rather than arbitrary recombination. The selected melody is transmitted as MIDI to RoboSax, a robotic saxophone mechanism that actuates the instrument's fingerings, while a human performer provides breath, tonguing, phrasing, dynamics, and tone color. This division of roles allows participants without instrumental training to influence the musical structure of a live performance while preserving the embodied expressiveness of acoustic wind performance. For SIGGRAPH Appy Hour, the work presents a participatory experience in which mobile interaction becomes immediately audible as live acoustic sound. By combining structurally coherent melody variation, audience-controlled interaction, and human-robot performance, RoboSax Melody Slot Machine explores how mobile music apps can extend beyond the screen and become part of a shared performative space.



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CCS Concepts

• Applied computing → Sound and music computing; • Human-centered computing → Interaction devices.

Keywords

Melody Slot Machine, RoboSax, GTTM, melody morphing, mobile music interaction, robotic musical instruments, participatory performance

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1 Introduction

Mobile music applications make musical creation accessible, but their output often remains within the screen, loudspeaker, or headphone. Live instrumental performance, by contrast, is embodied and situated: sound is shaped by the performer's breath, timing, articulation, and physical presence. RoboSax Melody Slot Machine connects these two domains by turning a participant's mobile interaction into live acoustic saxophone performance.

The system combines an iPad-based Melody Slot Machine interface for selecting melodic variations with RoboSax, a robotic saxophone mechanism that controls fingering while leaving breath

and articulation to a human performer. A participant selects melodic fragments through a slot-machine-like visual interface. The selected melody is transmitted as MIDI to RoboSax, whose actuators operate the saxophone keys. The performer produces the acoustic sound by blowing into the instrument and shaping tonguing, phrasing, dynamics, and tone color. Figure 1 shows the overall participatory workflow.

The melodic variations are grounded in the Generative Theory of Tonal Music (GTTM), which represents musical structure through hierarchical grouping, metrical, time-span, and prolongational structures [Hamanaka et al. 2006; Lerdahl and Jackendoff 1983]. Our previous work has developed GTTM analysis systems such as ATTA, FATTA, and the Interactive GTTM Analyzer [Hamanaka et al. 2005, 2007; Hamanaka and Tojo 2009], GTTM-based melody morphing [Hamanaka et al. 2008, 2022], sigmaGTTM III and deepGTTM-I/II [Hamanaka et al. 2015, 2016a,b], and a GTTM database [Hamanaka et al. 2014]. Melody Slot Machine translated these structurally grounded transformations into an interactive visual interface for selecting melodic variations [Hamanaka 2019; Hamanaka et al. 2019].

For SIGGRAPH Appy Hour, we frame RoboSax Melody Slot Machine as a mobile-to-acoustic performance experience. The contribution is not only the integration of an app and a robotic instrument, but also the design of a participatory performance loop in which audience members, a mobile interface, a robotic mechanism, and a human performer jointly shape the music.

2 Design Goals

RoboSax Melody Slot Machine was designed around three goals: structural playfulness, performative legibility, and embodied sound.

The first goal is structural playfulness. Participants should be able to play with melodic structure without composing note by note or learning a specialized music-editing interface. The slot-machine metaphor supports quick selection, comparison, and surprise, while GTTM-based melody morphing keeps the resulting phrases structurally related. In this way, the interface offers a form of guided musical freedom: the participant can make visible and immediate choices, but the system constrains those choices within a coherent melodic space.

The second goal is performative legibility. Appy Hour demonstrations must be understandable in a short encounter, often while other activities are taking place nearby. The system therefore makes each stage of the interaction visible: the participant manipulates the tablet interface, the selected melody is sent to RoboSax, the robotic mechanism actuates the saxophone keys, and the performer turns the mechanically prepared fingering into acoustic sound. The audience can follow the chain from visual selection to physical actuation and finally to live music.

The third goal is embodied sound. The system deliberately avoids replacing the performer with full automation. RoboSax automates fingering, but the human performer remains responsible for breath, tonguing, phrasing, dynamics, and tone color. This separation allows the robot to handle mechanical precision while the performer retains expressive agency. The result is a shared performance loop in which the participant, app, robot, and performer each contribute to the musical event.

3 Mobile Melody Selection

Melody Slot Machine provides a playful interface for exploring melodic variation. The interface displays musical staves on an iPad as slot-machine-like strips. Instead of composing note by note, participants browse a space of related phrases and select the version they wish to hear. The visual metaphor emphasizes play, chance, and immediate feedback, while the use of notation preserves a connection to the underlying musical material.

The melody space is generated through GTTM-based melody morphing. Rather than treating melodies as flat note sequences, the method represents each melody as a time-span tree, in which notes have different degrees of structural importance. Melody morphing links common events between two melodies, partially reduces branches that differ, and recombines the resulting structures to generate intermediate melodic forms [Hamanaka et al. 2008, 2022]. This structural view has been supported by GTTM analysis systems and datasets, including ATTA, FATTA, the Interactive GTTM Analyzer, and the GTTM database [Hamanaka et al. 2005, 2007, 2014; Hamanaka and Tojo 2009]. Figure 2 summarizes this process.

This approach is important for participatory performance. Non-expert users can make quick choices, but those choices remain within a coherent melodic space. The participant does not need to understand GTTM, notation, or composition theory; the interface presents musically meaningful alternatives as manipulable visual objects. Thus, Melody Slot Machine supports accessible musical agency without reducing performance to arbitrary sound triggering.

4 From Mobile App to Acoustic Performance

The selected melody is sent from Melody Slot Machine to RoboSax as MIDI. RoboSax builds on our RoboSax ensemble system with melody morphing [Koutaki and Hamanaka 2025] and is a semi-automatic robotic saxophone mechanism attached to a standard saxophone. A microcontroller receives MIDI note messages and drives actuators corresponding to saxophone fingerings. The mechanism operates the keys required for each pitch, while the human performer remains responsible for producing and shaping the sound.

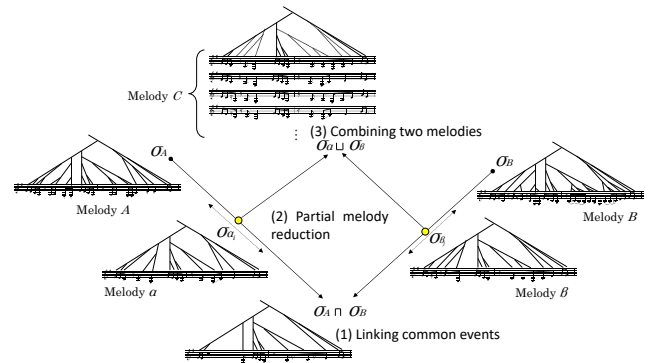


Figure 2: GTTM-based melody morphing, showing common-event linking, partial melody reduction, and melody combination.

The physical mechanism makes the transformation from symbolic melody to acoustic performance visible. RoboSax is mounted on a saxophone using a custom frame that holds servomotors, control circuits, and a microcontroller. The saxophone keys are connected to the servomotors by wires; when a servomotor rotates and pulls a wire, the corresponding key moves and closes the tone hole. Because the mechanism receives MIDI note messages from the tablet-based system, the same melody data selected through the interface can be routed directly to physical key actuation.

This design separates wind-instrument performance into complementary roles. RoboSax automates fingering, a major technical barrier for beginners and a demanding component of fast passages. The human performer provides breath, tonguing, dynamics, phrasing, and tone color. RoboSax is therefore not a fully autonomous musical robot that replaces the performer. It is a role-sharing instrument that reallocates technical and expressive responsibilities between mechanism and musician.

This role-sharing model is central to the Appy Hour experience. If the system only played synthesized sound, the participant's interaction would remain a conventional mobile music interaction. If the robot performed everything autonomously, the performer might disappear from the experience. RoboSax Melody Slot Machine instead creates a hybrid loop: the participant controls musical structure, the robot realizes the required fingering, and the performer gives the melody acoustic life through embodied wind performance.

5 Appy Hour Demonstration Experience

At SIGGRAPH Appy Hour, RoboSax Melody Slot Machine is presented as a participatory demonstration rather than a fixed concert piece. A participant interacts with the iPad interface, selects melodic fragments, and hears the resulting melody performed live by a human performer with RoboSax. Other attendees can observe both the selection process and the physical production of sound.

The demonstration follows a short interaction cycle. First, the participant views the tablet interface and chooses a melodic variation. Second, the system transmits the selected phrase to RoboSax as MIDI. Third, RoboSax actuates the saxophone keys while the performer supplies breath and articulation. Finally, the participant hears the selected melody as live acoustic sound. The cycle repeats, letting participants compare variations and reshape flow.

The experience emphasizes three forms of immediacy: the score-like strips and slot-machine metaphor make melodic variation visible; selections are heard as acoustic saxophone sound rather than as abstract data or delayed rendering; and the participant's choice is shared with surrounding attendees through the performer's embodied action. In this setting, the iPad is not merely a controller or composition tool but an entry point into collaborative performance.

The demonstration is also designed as a visual presentation. In addition to the live performer and RoboSax, attendees can view the tablet interface for melody selection. In some concert-format presentations, a camera view of the saxophone mechanism has been used to make the relationship between selected notes and robotic fingering visible, but Appy Hour does not use a camera. This presentation communicates the transformation chain: visual choice, symbolic melody, MIDI control, robotic fingering, breath, and acoustic sound.

6 Discussion and Conclusion

RoboSax Melody Slot Machine extends mobile music interaction into physical and social performance by combining GTTM-based melody generation, accessible mobile interaction, and robotic instrumental performance.

GTTM-based melody morphing constrains melody space so participant choices remain structurally coherent. The tablet interface makes these variations accessible, while RoboSax transforms selected melodies into physical fingering and the performer supplies expressive sound. Performance is distributed across participant, app, robot, and performer.

This distribution of agency is the core of the work: the participant chooses the melody, the model maintains structure, the robot executes fingering, and the performer shapes sound. It also reflects our semi-automatic wind-instrument approach, where separating fingering from breath lowers entry barriers while preserving expressive control. For Appy Hour, this is presented as a compact public experience in which audience members enter the musical loop through mobile melody selection.

The main contribution is an experience in which a mobile app leaves the screen and becomes audible, visible, and embodied. By connecting playful melody selection to live acoustic performance, the work invites attendees to become co-creators and demonstrates a new role for mobile interfaces in shared performance. The experience emphasizes shared authorship in live performance.

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